

Practical fairness-audit checklist under this protocol

Benchmark audit habits supported by the reported evidence (not a compliance checklist)

1. Audit more than one dataset

Dominant protected settings differed across Adult, UCI Bank, and ACSIncome.

2. Report component metrics, not only composites

DI can dominate CFS while |SPD| stays modest; read cells alongside marginals.

3. Match mitigation to objective

Equalized Odds and EG-DP cost more accuracy/runtime; Reweighing is lighter.

4. Treat EG as scoped unless fully evaluated

Main-grid Pareto uses baseline, Reweighing, and EO; EG overlays are illustrative

5. Check protocol choices

Report calibration, age binning, and other preprocessing decisions.

6. State runtime and accuracy budget

Mitigation scope and SAM-Fair selections depend on stated budgets.

7. Make selection assumptions explicit

Document ACFS weights, delta, and optional uncertainty/runtime penalties.